

# Aditya Kumar Sethia

Ambitious third-year B.Tech student with strong foundational computer science knowledge, combining generative modeling and high-performance computing experience to architect scalable, reliable, and efficient software solutions within a cooperative engineering environment.

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## TECHNICAL SKILLS

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**Languages:** Python, C/C++, Bash, SQL, Java

**CS:** Data Structures & Algorithms, Complexity Analysis, OOP Design, Operating Systems, DBMS/RDBMS, Distributed Systems, CI/CD, Unit Testing, Agile

**Frameworks:** PyTorch, PyTorch Lightning, LangChain, LangGraph, TensorFlow

**Tools & Systems:** Git/GitHub, FastAPI, Linux (Ubuntu), Flask, Streamlit

**Core Domains:** Deep Learning, Machine Learning, Generative AI, Data Science, Probabilistic Modelling, Parallel Computing

**Soft Skills:** Team Player, Time Management, Critical Thinking, Communication

## EDUCATION

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**Bachelor of Technology (B.Tech) in Artificial Intelligence & Data Science**

*Expected 2027*

IcfaiTech, IFHE Hyderabad | CGPA: 9.77

## EXPERIENCE

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**Open-Source Contributor (PyTorch Ecosystem)**

*January 2026 – Current*

neural-lam & Kornia

- Refactored neural-lam in Python, an object-oriented language, applying strict Object-Oriented Programming (OOP) paradigms, S.O.L.I.D. principles, and modular design to enable scalable, reliable probabilistic forecasting.
- Diagnosed and fixed a DistributedDataParallel initialization race in neural-lam (merged PR mllam #193), restoring correct multi-GPU distributed training under PyTorch Lightning; covered the fix with regression tests.
- Drove innovative optimizations to linear color transformations within the Kornia computer vision library, merging multiple core pull requests to accelerate processing efficiency and performance.

**Software Engineering Intern (Python & AI)**

*June 2025 – July 2025*

Eficens Software Services

- Engineered scalable software solutions — including a Retrieval-Augmented Generation (RAG) application with a secure FastAPI backend — leveraging LangChain, PyTorch, and ChromaDB for web content vectorization and LLM-based querying, while collaborating with Technical Program Managers and Product Managers to define requirements and uphold product quality, reliability, and performance.
- Collaborated within a cooperative team environment to author a comprehensive suite of 300+ Bash scripts automating Ubuntu 24.04 security audits, improving operational efficiency, consistency, and accountability while enforcing strict compliance with industry-standard CIS benchmarks.

## PROJECTS

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**Generative Physics Simulation (Diffusion)**

- Developed a PyTorch-based latent diffusion model — a cutting-edge technology in generative AI — to simulate complex fluid dynamics and chemical interactions by directly solving Partial Differential Equations (PDEs).
- Built an interactive JS/HTML visualization dashboard to render and analyze real-time inferences of the Shallow Water Equations, improving observability and performance monitoring.

**Plant Disease Detection (Computer Vision)**

- Trained and optimized a robust deep learning classifier in TensorFlow, engineering technical solutions to accurately identify multiclass crop diseases from raw agricultural imagery.
- Engineered efficient data augmentation and image preprocessing pipelines to significantly improve model generalization, reliability, and inference performance under varying visual conditions.